

Rearrangements in the Second Intron of the *RARA* Gene Are Present in a Large Majority of Patients With Acute Promyelocytic Leukemia and Are Used as Molecular Marker for Retinoic Acid-Induced Leukemic Cell Differentiation

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Chromosome 17 breakpoints in translocation t(15;17), a hallmark for acute promyelocytic leukemia (APL), have been shown to disrupt the retinoic acid receptor- α (*RARA*) gene. In this study, DNA probes around the second exon of the *RARA* gene showed rearrangements not previously detected. Analysis of 25 Chinese APL cases showed that *RARA* gene rearrangements were present in 23 cases (92%). The breakpoints were mapped unequivocally in 22 cases within the second intron of the gene. Therefore, the *RARA* gene rear-

angement provides us with a specific marker of the disease. Simultaneous molecular and cytologic studies showed that the *RARA* gene rearrangements persisted during the first 2 to 3 weeks of all-*trans* retinoic acid (ATRA) therapy when differentiated granulocytes predominated in bone marrow, while these rearrangements disappeared after achieving complete remission. These data indicate that ATRA induces differentiation of APL cells.

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THE CHROMOSOME translocation t(15;17)(q22;q12-21), a specific marker of acute promyelocytic leukemias (APL), has been shown to involve the retinoic acid receptor- α (*RARA*) gene on chromosome 17 and the *MYL* gene on chromosome 15.¹⁻⁶ The involvement of the *RARA* gene is of particular interest, as APL cells can be induced to differentiate upon treatment with the all-*trans* retinoic acid (ATRA).⁷⁻¹⁰ We have previously shown that the *RARA* gene was disrupted in its 5' part in 10 of 16 Chinese APL cases studied.⁵ On the other hand, the presence of abnormally sized *RARA* messenger RNA (mRNA) in one of the six cases without obvious genomic rearrangement suggested that the chromosome breakpoint was also located within the gene. Recently, the 5' end of the human *RARA* gene has been well defined: exon 1 encodes the 5' untranslatable region and exon 2 encodes the A region of *RARA*. The historically called intron 1 actually separates exon 2 from exon 3 coding for the B region of *RARA* and should be designated as intron 2.^{11,12} Using probes representative of the 5' extremity of the 17-kb second intron recently isolated in our laboratory, we have been able to detect gene rearrangements in four of six cases in which rearrangements could not be shown previously with a probe corresponding to the 3' end of the intron 2. Combined use of probes covering the entire intron 2 showed rearrangements in 23 of 25 APL cases, and most of these rearrangements

were mapped within this intron. Taking the *RARA* gene rearrangement as a molecular marker of the disease, we found that the APL cell population undergoes differentiation during remission induction with ATRA before the achievement of complete remission (CR).

MATERIALS AND METHODS

Samples. Bone marrow samples from 25 APL cases were studied. The clinical data for cases 1 to 16 were previously described,⁵ and correspond to the same case numbers in this report. Nine more recent cases (no. 17 to 25) were included in this work (Table 1). Diagnosis was made according to the French-American-British (FAB) criteria.¹³ Twenty-four cases were studied at the diagnosis, one case at relapse (Table 1). In two cases (no. 20 and 21), bone marrow samples were analyzed during the remission induction with ATRA and after achievement of CR. In case 3, bone marrow cells were also analyzed during CR. Approval was obtained from the Institutional Review Board for these studies. Informed consent was provided according to the Declaration of Helsinki.

Cytogenetics. Chromosome analysis performed in 20 cases according to the standard G and R banding procedures¹⁴ showed the presence of the standard chromosome translocation t(15;17) in 19 cases. A translocation t(11;17) was found in the remaining case.

Molecular analysis. Southern blot experiments were performed using standard protocols.¹⁵ The intensities of normal bands, as well as rearranged bands, were measured on a densitometer (Shimadzu CS-930; Shimadzu, Kyoto, Japan) in cases 20 and 21 during ATRA induction of CR. For studying the human *RARA* gene structure, a human genomic DNA library was screened with a cDNA probe containing partial sequences from exons 2 and 3 of the *RARA* gene.⁵ Seven phage clones were obtained and their restriction map is shown in Fig 1. Relevant DNA fragments devoid of repetitive sequences were subcloned into the pGEM blue plasmid (Promega, Madison, WI) to generate probes. The positions of the five probes used in this study are indicated in Fig 1. Probes were labeled with the random primer system to obtain a specific activity of approximately 1×10^9 cpm/ μ g DNA.

RESULTS

Generation of new probes around the second exon of the *RARA* gene allows detection of rearrangements in most APL patients. In a previous study,⁵ we described a *RARA* gene region that covers a domain from the 3' portion of the second intron to the last exons (see Fig 1, region composed of six overlapping phage clones). Among the previously studied 16 APL cases, rearranged *RARA* gene was detected

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Table 1. Results of Cytogenetic and Molecular Studies in 25 Cases of APL

Patient No.	Sex/Age (yr)	Stage	t (15; 17)	Rearrangement of <i>RARA</i> Gene		Response to ATRA
				5' (0.6-kb HE) Probe	3' (5.5-kb EE) Probe	
1	M/13	D	+	+	+	No
2	M/14	D	ND	-	+	*
3	M/12	D	+	+	+	CR
		CR	-	-	-	
4	F/40	Relapse	+	-	-	CR
5	M/42	D	+	+	+	*
6	F/57	D	+	+	+	CR
7	F/29	D	+	+	-	CR
8	M/29	D	+	+	-	CR
9	M/33	D	ND	ND	+	CR
10	F/26	D	+	+	-	CR
11	M/28	D	ND	ND	+	*
12	M/35	D	+	+	-	*
13	F/29	D	+	-	-	CR
14	M/18	D	+	-	+	CR
15	F/6	D	+	+	+	CR
16	F/11	D	+	+	+	CR
17	F/41	D	+	+	+	CR
18	F/5	D	+	+	+	CR
19	F/29	D	+	+	+	CR
20	F/34	D	+	-	+	CR
		CR	-	-	-	
21	M/27	D	+	+	-	CR
		CR	-	-	-	
22	M/36	D	ND	+	+	CR
23	M/67	D	t (11; 17)	-	+	No
24	F/11	D	ND	+	+	CR
25	M/28	D	ND	+	+	CR

Abbreviations: D, at diagnosis; ND, not done.

Cytogenetic and molecular analyses were performed with bone marrow samples in all cases studied.

*Early death (within the first 10 days of ATRA).

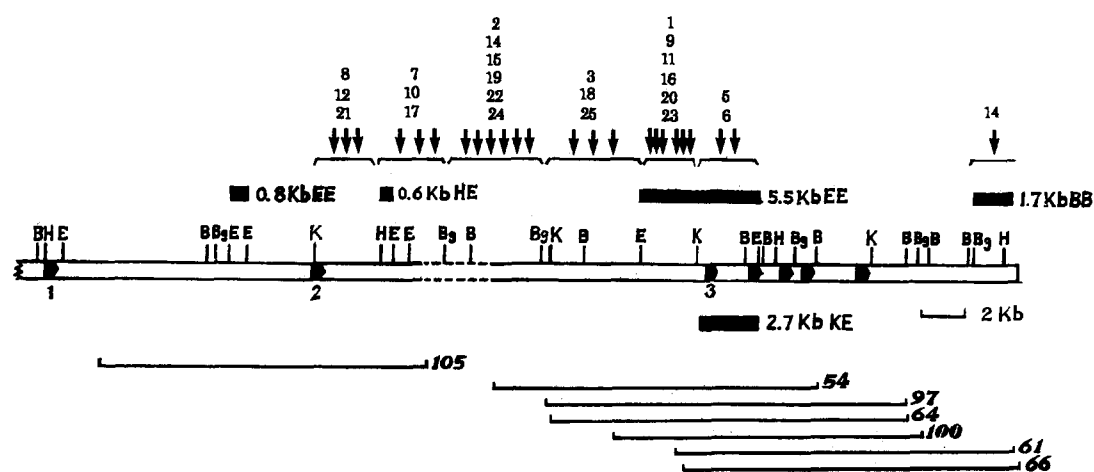


Fig 1. Restriction map of the human *RARA* gene according to the literature and our own results^{2,5,11,12} and localization of the chromosome 17 breakpoints in APL patients. Abbreviations for restriction enzymes: B, *Bam*HI; Bg, *Bgl* II; E, *Eco*RI; H, *Hind*III; K, *Kpn* I. Arrowed boxes represent exons and their transcriptional orientations. Dotted lines correspond to the gap between the clone 105 and other phage clones. A *Bam*HI and a *Bgl* II site were mapped in this region according to results of Southern experiments with human genomic DNA. Black bars above and below the map indicate positions of probes used in this study (see also text). Breakpoints in 23 APL cases were mapped and their positions arrowed. The corresponding case numbers are shown above the arrows. Case 14 also had a rearrangement in the gene's 3' end.

in 10 cases by the 5.5-kb *EcoRI-EcoRI* (EE) probe, which is situated at the 3' end of the intron 2. In the six remaining cases, no obvious genomic rearrangement was observed.

To see whether the chromosome 17 breakpoints took place in a more 5' region of the gene, we analyzed in detail clone 105 (Fig 1), which was mapped to the most 5' part of our cloned *RARA* gene, but did not overlap the other phage clones. To bridge the gap, we generated a *HindIII-EcoRI* probe from the clone 105 (the 0.6-kb HE probe, Fig 1) and performed hybridization tests with total genomic DNA from control samples and some APL patients. This probe detected the same 18-kb *HindIII* band on Southern blots as the 5.5-kb EE probe. Moreover, in *HindIII* DNA digests, it detected one of the two rearrangements shown by the 5.5-kb EE probe (Fig 2). These results allowed establishment of the transcriptional orientation of the DNA region contained in the clone 105, as well as the size of the gap (3 kb) between clone 105 and other clones.

The 0.6-kb HE probe and a 0.8-kb EE probe localized 3 kb upstream of the second exon were used to further study the *RARA* gene structure in the six cases in which rearrangements could not be detected previously (Fig 3). In addition to the germline bands, rearranged bands were detected with the HE probe in four of the six cases. Theoretically, the rearrangements in cases 7 and 10 should also be detected by the 5.5-kb EE probe with *HindIII* enzyme. However, the size of the rearranged bands hybridizing with this probe may be close to that of the large germline *HindIII* band (18 kb) and, therefore, produce a comigration. In contrast, the reciprocal rearranged bands were of smaller size and were detected by the HE probe. Hybridization with the 0.8-kb EE probe confirmed the rearrangement in cases 8 and 12. No rearrangements were observed in cases 4 and 13 with the panel of probes covering the entire *RARA* gene.

Recently, we observed nine new APL cases (cases 17 to 25). A gene rearrangement was detected in all these cases, including case 23 with a variant translocation t(11;17). Taken together, we were able to demonstrate the presence of a *RARA* gene structural abnormality in 23 of 25 cases (92%) studied.

Mapping of the chromosome 17 breakpoints within the *RARA* gene in 23 APL cases. The structure of the genomic *RARA* gene, according to the recent data,^{6,11,12} shows that the gene's first and second introns are 12 and 17 kb long, respectively. Availability of probes for both extremities of

intron 2 made it possible to precisely localize the APL chromosome breakpoints previously shown to be located in the 5' region of the *RARA* gene. We also took advantage of the presence of two *Kpn* I sites within the second exon and just upstream of the third exon, respectively, to map the breakpoints with respect to the exon-intron structure of the gene.

Southern experiments were subsequently performed in relevant cases with *EcoRI*, *Bgl*II, *Hind*III, and *Kpn* I restriction enzymes and the 0.6-kb HE probe, as well as the 2.7-kb *Kpn* I-*EcoRI* (KE) probe (see Fig 1). As shown in Fig 1 (arrows), among our 23 APL cases with rearranged *RARA* gene, the breakpoints were mapped in the second intron in 21 cases. In the remaining two cases (no. 5 and 6), the breakpoints were localized downstream of the *Kpn* I site 5' to the third exon. However, because a small intron 2 segment is 3' to the *Kpn* I site, the localization of those two breakpoints within the second intron cannot be excluded. Recently, one chromosome breakpoint in case 5 was cloned in our laboratory and it has been shown to lie at the 3' end of the second intron (J.H. Tong and Z. Chen, unpublished data). It is also noticeable that the chromosome breaks were evenly distributed within the second intron without obvious clustering in a subregion.

***RARA* gene rearrangement as a molecular marker for leukemic cell differentiation.** Because ATRA has been shown to induce morphologic differentiation of APL cells, it was of interest to follow in parallel the evolution of the *RARA* molecular rearrangements during ATRA treatment. This study was performed on bone marrow samples collected during the remission induction by ATRA in cases 20 and 21. The results in case 20 are shown in Fig 4. While the bone marrow promyelocyte percentage decreased to 4% (this percentage being close to the limit of sensitivity of Southern analysis to detect the minimal residual disease in our laboratory) on day 14 of ATRA treatment, there was a remarkable increase of differentiated granulocytes (myelocytes, metamyelocytes, band forms, and polymorphonuclear granulocytes, which accounted for 80% of the bone marrow nucleated cells) exhibiting some morphologic anomalies such as unbalanced nuclear/cytoplasmic maturation. Interestingly, the Southern analysis performed meanwhile showed the same *RARA* gene rearrangement as before ATRA. Importantly, the intensities of the rearranged bands measured on densitometer were approximately iden-

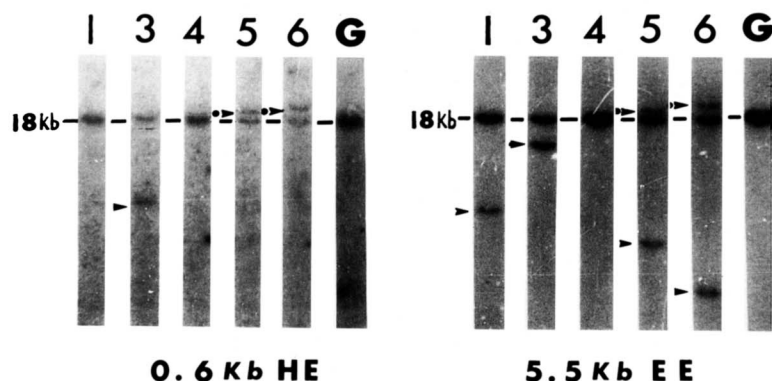


Fig 2. Comparison of Southern hybridization results with probes for 5' (0.6-kb HE) and 3' (5.5-kb EE) extremities of the *RARA* gene's second intron in APL patients (numbered lanes) and normal control (G). Rearranged bands are arrowed; those with a black point indicate rearrangements shown by both probes.

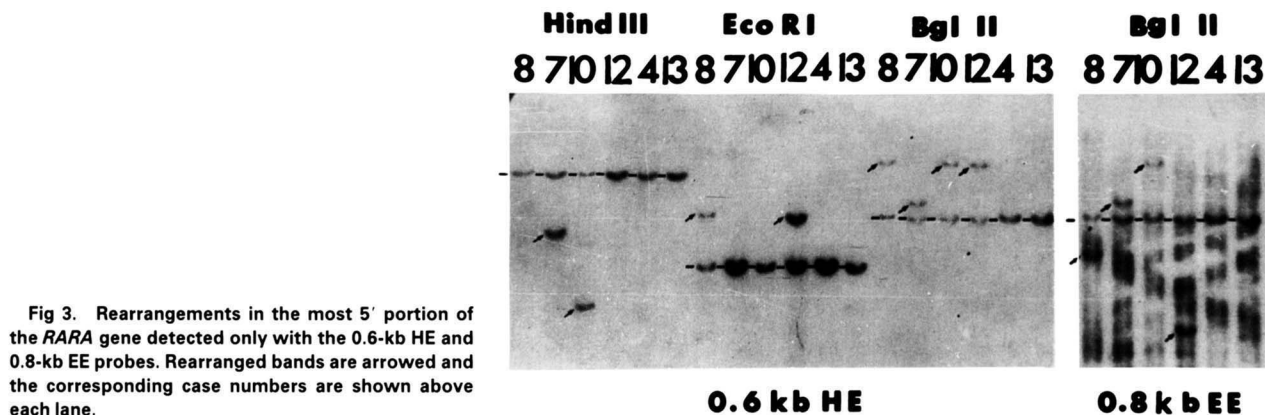


Fig 3. Rearrangements in the most 5' portion of the *RARA* gene detected only with the 0.6-kb HE and 0.8-kb EE probes. Rearranged bands are arrowed and the corresponding case numbers are shown above each lane.

tical in the untreated and ATRA-treated samples (see Fig 4A), excluding the possibility that rearranged fragments in the ATRA-treated cells only originated from residual leukemic promyelocytes. This finding strongly suggests that the morphologically differentiated granulocytes were carrying the molecular marker of the leukemic clone. The normal *RARA* gene configuration reappeared after 28 days of ATRA, when CR was achieved.

Results in case 21 were similar (Fig 5). Again, the percentage of bone marrow promyelocytes was significantly reduced (20%) on day 21 of ATRA treatment, with concomitant augmentation of mature myeloid cell count (66%). Densitometric tracing showed that the intensities of the rearranged band (~42%) were much higher than the expected value (10%) for 20% of leukemic promyelocytes. On the other hand, the strong intensities of rearranged

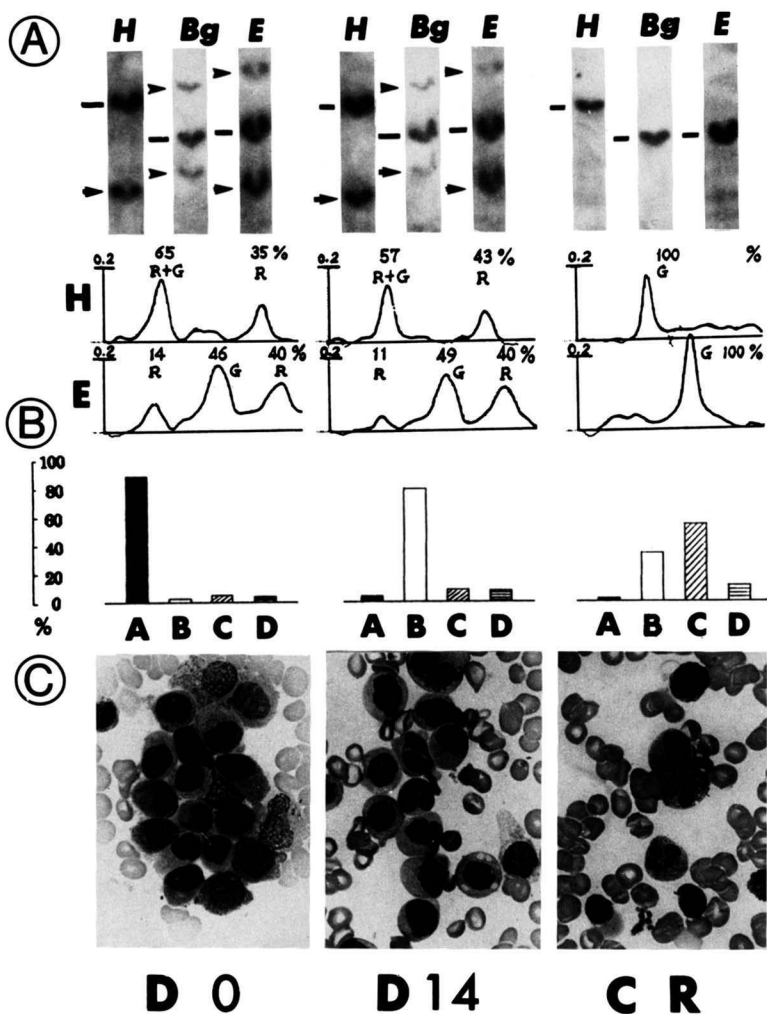


Fig 4. Simultaneous molecular and cytologic study in case 20 during the remission induction with ATRA. (A) Southern analysis using the 5.5-kb EE probe performed on day 0, day 14, and during CR after 28 days of ATRA therapy. Abbreviations for restriction enzymes: H, *HindIII*; Bg, *Bgl II*; E, *EcoRI*. Black bars indicate normal bands. Rearranged bands are arrowed. The intensities of germline (G) and rearranged (R) bands in *HindIII* (H) and *EcoRI* (E) digests were determined using a densitometer (Shimadzu, CS-930) and are represented as percentages of the area occupied by each peak. (B) Schematic representation of the percentages of bone marrow nucleated cells: promyelocytes (A), differentiated myeloid cells (B, including myelocytes, metamyelocytes, band forms and polymorphonuclear granulocytes), erythroid cells (C), and other cells (D) on the same day as the Southern analysis. (C) Corresponding bone marrow morphological changes during ATRA treatment.

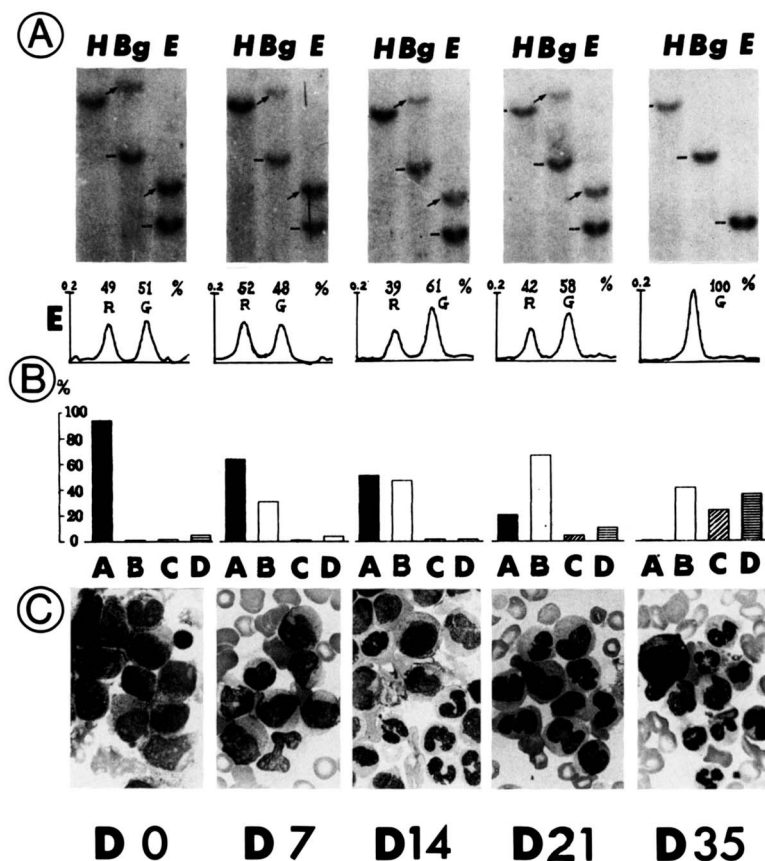


Fig 5. Molecular and morphological study in case 21 during ATRA treatment. (A) Southern analysis using the 0.6-kb HE probe before (day 0), and on days 7, 14, 21, and 35 of ATRA treatment, and the densitometric tracing of germline (G) and rearranged (R) bands in *EcoRI* digests. (B) Columns representing the differential count (%) of bone marrow nucleated cells: promyelocytes (A), differentiated myeloid cells (B), erythroid cells (C), and other lineage cells (D) analyzed on the same days as the Southern analysis. (C) Corresponding bone marrow pictures.

bands are consistent with the sum of the percentages of promyelocytes and differentiated myeloid cells (86%) in bone marrow. The abnormal bands disappeared on day 35, when the bone marrow was repopulated with morphologically normal cells (Fig 5).

DISCUSSION

Cytogenetic studies have shown that practically all APL patients have the t(15;17) when investigated with optimal techniques.¹⁷⁻¹⁹ Using the in situ hybridization technique, the breakpoint on chromosome 17 was recently mapped to the region 17q21 instead of 17q11-12 as previously reported.²⁰ This localization is consistent with the reported localization of *RARA* gene,²¹ which has been shown to be disrupted by the chromosome 17 breakpoints. In the present work on 25 fresh cases of APL, we showed that the rearrangement of the *RARA* gene resulting from the t(15;17) was found in 23 of 25 APL patients. It may be interesting to note that the *RARA* gene was rearranged in case 23 with a variant translocation t(11;17), suggesting that common molecular basis exists in classical t(15;17) and its variants. A study of *RARA* gene structure in 20 samples of other leukemia types (chronic myelogenous, acute lymphatic, acute myelogenous-M1) did not show any rearrangement (not shown). Therefore, the *RARA* gene rearrangement should be considered as a specific marker of APL. This questions the definition of the two cases without *RARA*

gene rearrangement: both cases had typical morphology of APL, both had the t(15;17), both had responded to ATRA therapy. These data support the notion that the breakpoint on chromosome 17 is located in the *RARA* gene locus, although not within the gene sequence covered by our probes. Further molecular study should be performed to get insight into the chromosome breakpoints in these cases.

One of the aims of this study was to highlight the precise localization of the chromosome 17 breaks in the *RARA* gene, because the variation of the breakpoints in terms of the exon-intron structure may result in different fusion genes with possibly functional differences. Using probes covering the whole *RARA* gene and appropriate restriction enzymes, we showed that the chromosome 17 breaks were evenly distributed in the second intron in almost all cases studied. This finding is somewhat different from a recent report by Borrow et al,² in which the breakpoints in their APL patients were more restricted in a region of 6 kb within the second intron. No obvious differences in clinical parameters, especially responsiveness to ATRA, could be observed between cases with chromosome 17 breakpoints localized at the 3' end of the second intron and those with breakpoints at the 5' end of this intron (Table 1). Because intron sequences are to be removed during the mRNA splicing process, variant chromosome breaks localized in the same intron would not change the size of the gene transcripts. As recently stressed,²¹ several chromosomal

translocations of acute leukemias are characterized by breakpoints localized within specific intron regions of genes producing chimeric proteins. In APL, this situation is confirmed for breakpoints involving the *RARA* gene. We recently found that rearrangements of the *MYL* gene on chromosome 15 were also characterized by their clustering in a limited intron region. However, some variant breakpoints localized in a different intron have been noticed, which may be the basis for the two types of abnormal fusion mRNA to be produced in our APL cases as previously reported.⁵

Previously described evidence suggesting that ATRA stimulates leukemic cell differentiation includes maturation of leukemic cell lines such as HL-60 upon effect of ATRA in vitro, abolition of the bone marrow suppression during the in vivo treatment, and appearance of differentiated myeloid cells carrying morphologic abnormalities such as Auer's rod after ATRA treatment. This latter observation was generally accepted as the strongest argument for differentiation induction effect of ATRA in APL patients.²³

In the present work, using the *RARA* gene rearrangement as molecular marker, we showed that, during the first 2 to 3 weeks of ATRA therapy, the majority of the differentiated bone marrow myeloid cells are derived from the leukemic clone. This finding may be relevant to the rare situation in which apparent clinical CR judged by a low percentage of bone marrow promyelocytes and restoration of peripheral blood counts (hemoglobin, white blood cells, and platelets) was observed by clinicians, while cytogenetic study showed still some t(15;17)-bearing mitosis in bone marrow. However, the t(15;17) generally disappeared when further examined after prolonged ATRA treatment. Therefore, our data add new evidence that ATRA reduces the leukemic cell mass in vivo by triggering their maturation. The molecular mechanism of the differentiating effect of ATRA remains to be elucidated.

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